l recording

successfully

proper connect.

Exp No:- 4

Date:- 6/8/2026

Aim:-

To implement port sniffing using raw sockets in Python and capture network packets passing through the network interface.

Objective:-

To understand how raw sockets work & how network packets are captured & analysis using Python programming.

Theory:-

- Packet is the process of monitoring & data process across a network.
- A packet sniffing intercepts packet and analyze their contents.
- Raw sockets provide access to lower network protocols, allowing applications to capture packets before they are passed by the operating system.
- In python, the socket module can be used to create raw sockets for packet capture system.

Algorithm:

1) Import the required socket module
2) Create a raw socket.
3) Bind the socket to the network interface
4) Receive Packets using the recvfrom() method.
5) Display the captured packet information.
6) Repeat the process continuously to monitor network traffic.

Program:-

```
Sniffer = Socket.Socket(Socket.AF_INET,
                        Socket.SOCK_RAW,
                        Socket.IPPROTO_IP)
host = Socket.gethostbyname(socket.gethostname())
Sniffer.bind((host, 0))
Sniffer.setsockopt(Socket.IPPROTO_IP, Socket.IP_HDRINCL, 1)
while True:
    raw_data, addr = Sniffer.recvfrom(65535)
    print('Packet received from:', addr)
    print(raw_data)
```

o/p:-

Packet received from
('192.168.1.5', 0)
b'E\x00\x00\x00...'

Packet received from:
('192.168.1.10', 0)
b'E\x00\x00\x00...

Result :-

Thus, Packet sniffer has successfully implemented using raw sockets in python & displayed packet were captured & displayed.